Ordinary classes and the three-valuation in the fifty-three factor

E.124. A saturated ordinary-unit basis, the next full class group and the exact bounded valuation image [Proved]

Keep the compatible roots and actual ideal maps of E.123. Write

$$\begin{aligned} &L_d = \mathbb{Q}(\beta_d), \quad \beta_d^{3^d} = 53\zeta_3, \quad \beta_{d+1}^3 = \beta_d, \\ &D_d = \mathrm{Cl}(L_d)\{3\}, \quad C_d = \mathrm{Cl}_S(L_d)\{3\}, \quad E_d = \mathcal{O}_{L_d}^\times, \qquad S \mid \{3, 53\}, \\ &q_d = [\mathfrak{q}_{3,d}], \quad t_d = [\mathfrak{q}_{53,d}], \quad N_d : D_{d+1} \longrightarrow D_d, \quad j_d : D_d \longrightarrow D_{d+1}. \end{aligned} \tag{E124.1}$$

Here N_d is ideal norm and j_d is ideal extension. The prime over fifty-three is the principal ideal (β_d). The eight literal ordinary units $\epsilon_1, \ldots, \epsilon_8$ in the first eight columns of E.122's upper witness list, together with ζ_3, give the following isomorphism after completion for multiplication by 3^n:

$$\mu_3 \times \mathbb{Z}_3^8 \xrightarrow{\sim} \widehat{E}_{2,3}, \qquad (z, a_1, \ldots, a_8) \longmapsto z \prod_i \epsilon_i^{a_i}. \tag{E124.2}$$

The inverse is the exponent map in this proved basis; its reduction modulo three is the explicit inverse residue matrix in the certificate. Every ordinary unit of L_2 is a field norm from L_3.

In E.123's actual three-ideal basis and the Smith basis below, the entire ordinary three-class groups and actual maps are

$$\begin{aligned} D_2 &= C_9 \oplus C_9 \oplus C_3, \qquad D_3 = C_{27} \oplus C_9 \oplus C_3, \\ q_2 &= (6, 3, 1), \qquad q_3 = (21, 0, 1), \\ N_2 &= \begin{pmatrix} 8 & 1 & 0 \\ 1 & 0 & 0 \\ 2 & 1 & 1 \end{pmatrix}, \qquad j_2 = \begin{pmatrix} 0 & 3 & 0 \\ 3 & 3 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \end{aligned} \tag{E124.3}$$

Each row is reduced modulo its target order. Thus q_2 has order three and q_3 has order nine. The actual automorphism $\sigma_2(\beta_3) = \zeta_3\beta_3$ acts identically on D_3. The full kernels and quotient maps are

$$\begin{aligned} &\ker N_2 = \langle (9,0,0) \rangle = \langle 3q_3 \rangle, \qquad \ker j_2 = \langle (3,0,0), (0,0,1) \rangle, \\ &N_2 j_2 = 3I_{D_2}, \qquad j_2 N_2 = 3I_{D_3}, \\ &P_2 = \begin{pmatrix} 7 & 7 & 0 \\ 2 & 4 & 3 \end{pmatrix}, \qquad P_3 = P_2 N_2 = \begin{pmatrix} 0 & 7 & 0 \\ 8 & 5 & 3 \end{pmatrix} \pmod 9, \\ &P_i : D_i \twoheadrightarrow C_2 = C_9 c_{31} \oplus C_9 c_{103}, \qquad \ker P_i = \langle q_i \rangle \quad (i = 2, 3). \end{aligned} \tag{E124.4}$$

For P_3, the identification $C_3 \to C_2$ is E.123's actual norm isomorphism. The subscripts $31, 103$ denote its literal pivot primes.

The ordinary-unit norm defect and completed kernel are

$$\begin{aligned} &E_2/N_{L_3/L_2}E_3 \xleftarrow{\sim} D_3/(3D_3 + \langle q_3 \rangle) \xrightarrow{\sim} C_3^2, \\ &[(r,s,t)] \longmapsto (r \bmod 3, s \bmod 3), \qquad \ker(\widehat{E}_{3,3} \to \widehat{E}_{2,3}) \simeq \mu_3 \oplus \mathbb{Z}_3^{18}. \end{aligned} \tag{E124.5}$$

The first arrow is the actual Hilbert–90 ideal-to-unit map, with the inverse below. Its defining norm subgroup is retained; no numerical two-row character on the eight-unit basis is assigned. The torsion factor in the kernel is the literal root subgroup.

Define actual integers a_d by $\mathrm{ord}(q_d) = 3^{a_d}$. Then

$$\begin{aligned} &a_2 = 1, \quad a_3 = 2, \qquad a_d \le a_{d+1} \le a_d + 1, \qquad h := \sup_{d \ge 2} a_d < \infty, \\ &D_d \simeq C_{3^{a_d+1}} \oplus C_9 \oplus C_3 \quad (d \ge 3), \qquad \varprojlim_{d, N_d} D_d \simeq C_{3^{h+1}} \oplus C_9 \oplus C_3. \end{aligned} \tag{E124.6}$$

The proof gives actual compatible basis choices and inverse maps. In those bases $q_d = (3, 0, 1)$; for $d \geq 3$ each norm reduces the first coordinate and preserves the other two. Extension multiplies any norm lift by three; its ambiguity is killed by three. The finite integer $h \geq 2$ is not numerically evaluated here. Without any basis choice, for every $d \geq 2$,

$$\begin{gathered} N_d \text{ is onto}, \quad \ker N_d \subset \langle q_{d+1} \rangle, \quad |\ker N_d| = 3^{a_{d+1}-a_d} \leq 3, \\ (\sigma_d - 1)^2 D_{d+1} = 3(\sigma_d - 1) D_{d+1} = 0, \qquad j_d N_d = 3 I_{D_{d+1}}, \quad N_d j_d = 3 I_{D_d}, \\ E_d / N_d E_{d+1} \simeq C_3^{3-a_{d+1}+a_d}, \qquad \ker(\widehat{E}_{d+1,3} \to \widehat{E}_{d,3}) \simeq \mu_3 \oplus \mathbb{Z}_3^{2 \cdot 3^d}. \end{gathered} \tag{E124.7}$$

The finite unit quotient keeps its actual norm subgroup. The proof supplies its Hilbert–90 comparison as well as its order.

Let $\mathcal{V}$ be E.123's coherent completed S-unit group and $\mathcal{E} = \varprojlim_{d,N} \widehat{E}_{d,3}$. For its literal valuation maps one has

$$\begin{gathered} \nu_3(\mathcal{V}) = 3^h \mathbb{Z}_3, \qquad (\nu_3, \nu_{53})(\mathcal{V}) = 3^h \mathbb{Z}_3 \times 2\mathbb{Z}_3. \\ \text{Choose } \gamma \in \mathcal{V} \text{ with } (\nu_3, \nu_{53})(\gamma) = (3^h, 0), \qquad \boldsymbol{\beta} = (\beta_d)_d. \\ \mathcal{E} \times \mathbb{Z}_3^2 \xrightarrow{\sim} \mathcal{V}, \qquad (u, k, t) \longmapsto u \gamma^k \boldsymbol{\beta}^t, \\ y \longmapsto \left(y \gamma^{-\nu_3(y)/3^h} \boldsymbol{\beta}^{-\nu_{53}(y)/2}, \frac{\nu_3(y)}{3^h}, \frac{\nu_{53}(y)}{2} \right). \end{gathered} \tag{E124.8}$$

All maps are continuous and both compositions are identities. Their literal rational norm is

$$N_{\mathbb{Q}}(u \gamma^k \boldsymbol{\beta}^t) = 3^{\,3^h k} 53^{\,2t}. \tag{E124.9}$$

Together with E.123's fixed divisor splitting s, the entire positional factor has the explicit coordinates

$$(u, k, t, v) \longmapsto u \gamma^k \boldsymbol{\beta}^t s(v), \qquad N_{\mathbb{Q}} = 3^{\,3^h k + \rho(v)} 53^{\,2t + \eta(v)} \prod_{p \in \mathcal{P}} p^{v_p}. \tag{E124.10}$$

The inverse takes the actual outside-S divisor v, then applies (E124.8) to $x/s(v)$. Replacing γ by $e\gamma$, $e \in \mathcal{E}$, changes u to ue^{-k} and preserves the valuation coordinates. These are E.46's actual total-norm coordinates for this positional factor. The remaining numerical invariant is h; the other factors and intrinsic lifting maps retain their definitions in E.23. The manuscript anchor is §51.16, fols. 733v–736v, scans172–178, through E.46's root-cubing and Kummer–Shapiro maps.

Proof. Literal units and saturation. The field L_2 has degree eighteen, is totally imaginary and has ordinary unit rank eight. Its three-primary roots of unity are exactly μ_3, by E.123's horizontal root calculation. The rational coefficient vectors of the eight ϵ_i are retained in the immutable witness input. E.122's independent relative-ring identities and complete principal divisors show that they are ordinary units. No regulator or upper class-number output is used for this assertion.

At a degree-one prime with residues $\beta_2 = b$, $\zeta_3 = z$, use the actual character $u \mapsto \log_z(\overline{u}^{(p-1)/3})$. The new certificate selects the nine primes

p	7	7	7	31	31	31	43	79	97
b	1	2	4	11	24	27	4	41	44
z	2	2	2	25	25	25	36	55	35

. (E124.11)

They are actual maximal-order primes: these roots of $X^{18} + 53X^9 + 53^2$ are simple, and the primes avoid its discriminant support $\{3, 53\}$. On the columns $(\zeta_3, \epsilon_1, \ldots, \epsilon_8)$ the residue

matrix is

$$F = \begin{pmatrix} 2 & 0 & 0 & 2 & 0 & 1 & 2 & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 & 0 & 1 & 2 & 2 \\ 2 & 0 & 0 & 2 & 1 & 1 & 2 & 1 & 1 \\ 1 & 2 & 1 & 0 & 2 & 0 & 0 & 2 & 1 \\ 1 & 2 & 1 & 0 & 0 & 1 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 2 & 2 & 0 & 1 & 0 \\ 2 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 \\ 2 & 0 & 2 & 0 & 2 & 0 & 2 & 0 & 1 \\ 2 & 0 & 1 & 1 & 1 & 1 & 2 & 0 & 1 \end{pmatrix} \pmod 3. \tag{E124.12}$$

The certificate retains F^{-1} and verifies both products. Since E_2/E_2^3 has dimension nine, these nine classes are its full basis. Their subgroup has rank eight modulo torsion and index prime to three. Completion gives (E124.2). One inverse procedure reads the exponents modulo three with F^{-1}, divides by that literal unit product, and takes the now existing cube in E_2; iteration recovers the exponent digits.

At the unique fifty-three place the residue field is $\mathbb{F}_{53}[z]/(z^2+z+1)$ with the specified reduction of ζ_3. The positive Artin character for L_3/L_2 on units is

$$\chi_{53}(u) = -\log_z(\overline{u}^{936}) \pmod 3, \qquad 936 = (53^2-1)/3. \tag{E124.13}$$

This is the tame symbol of (β_2, u) in E.119's arithmetic Artin convention. Its value on every basis element in (E124.2) is zero, by direct finite-field evaluation. Hence all ordinary units are local norms there. At an unramified place every unit is a norm, and the product of the local Artin characters makes the character at the sole three-place zero. There are no real places. Milne VIII Theorem3.1, with its injective Tate-cohomology proof, now gives $E_2 \subset N_{L_3/L_2} L_3^{\times}$ [1]. The local-to-global product is V Proposition5.2 and Theorem5.3; V Example5.4 states its Hilbert-symbol form.

The two full ordinary class groups. E.119 gives $\mathrm{Cl}(L_1)\{3\} = C_9 \oplus C_3$ and $\mathrm{Cl}_S(L_1)\{3\} = C_3^2$. Because $(\beta_1) = \mathfrak{q}_{53,1}$, their kernel is generated by q_1 and has order three. E.123's literal horizontal norm w has complete divisor $3\mathfrak{q}_{3,2} - \mathfrak{q}_{53,2}$. Thus $w\beta_2$ has divisor $3\mathfrak{q}_{3,2}$, while $N_1 q_2 = q_1$ proves that q_2 has exact order three. Removing the S-primes gives the actual exact sequence

$$0 \longrightarrow \langle q_d \rangle \longrightarrow D_d \longrightarrow C_d \longrightarrow 0. \tag{E124.14}$$

The class q_d is three-primary: extensions of the relation for q_2 annihilate it by powers of three. The other removed class is zero, so there is no suppressed prime-to-three term.

At level two, $|C_2| = 81$ gives $|D_2| = 243$. E.123's literal ideals I_1, I_2, I_3 have complete principal power relations of orders nine, nine and three. Their classes generate C_2, and their combination $(6, 3, 1)$ is q_2. They generate D_2; the resulting surjection from $C_9 \oplus C_9 \oplus C_3$ is an isomorphism by equal orders. This proves its actual ideal basis in (E124.3).

The cyclic cubic L_3/L_2 has exactly two totally ramified finite places and no infinite ramification. The field-unit norm index just proved is one. The three-primary ambiguous class number formula therefore gives

$$|D_3^{\sigma_2}| = |D_2|\, 3^{2-1} = 729. \tag{E124.15}$$

This is Lemmermeyer Theorem1(1), proved by its fixed ideal calculation and the unit Herbrand quotient [2]. Extension of $3q_2 = 0$ gives $9q_3 = 0$. Equation (E124.14) and $|C_3| = 81$ bound $|D_3|$ by 729. Equality proves that q_3 has order nine and that σ_2 fixes the entire ordinary group.

The actual idelic presentation and local signs. Let H_3 be the ordinary Hilbert three-class field of L_3. It is Galois over L_2. The group $\Gamma = \mathrm{Gal}(H_3/L_2)$ is an extension of D_3 by a cyclic

group with trivial action. Choosing a lift of that generator proves that Γ is abelian, of order 2187.

The two local unit norm quotients have order three; denote their positive generators by e_3, e_{53}. The literal norm uniformizers are $1+\beta_2 = N(1+\beta_3)$ and $\beta_2 = N\beta_3$. They fix the local valuation splittings and the zero uniformizer characters. The other three generators are actual lifts of I_1, I_2, I_3. Complete factorization in the certified maximal order gives their Frobenius characters $(1,0,2)$ in L_3/L_2.

Specifically, I_1 is the prime above 2683 with residues $(\beta_2, \zeta_3) = (1567, 2046)$; I_2 is the prime above 7 with residues $(1,2)$. The ideal I_3 is the product of that prime, the prime above 31 with residues $(24,25)$, and the stated prime above 2683. Their factor characters are 1, 0, and $(0,1,1)$ respectively. The nf-only program checks these ideal factorizations and every residue; in particular it does not treat I_3 as prime.

For the retained principal generators a_i of I_1^9, I_2^9, I_3^3, direct use of (E124.13) gives

$$\begin{array}{c|ccc} & a_1 & a_2 & a_3 \\ \hline \chi_3 & 0 & 2 & 0 \\ \chi_{53} & 0 & 1 & 0 \end{array}. \tag{E124.16}$$

Their residue pairs in $\mathbb{F}_{53}[z]/(z^2+z+1)$ are $(8,16), (43,36), (0,45)$. The three-place row follows from reciprocity: every outside-S exponent in these complete principal divisors is divisible by three. All global ordinary unit characters vanish. The exact idelic presentation is consequently

$$R = \begin{pmatrix} 3 & 0 & 0 & 2 & 0 \\ 0 & 3 & 0 & 1 & 0 \\ 0 & 0 & 9 & 0 & 0 \\ 0 & 0 & 0 & 9 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{pmatrix}, \qquad \Gamma = \mathbb{Z}_3^5/R\mathbb{Z}_3^5, \qquad \phi = (1,1,1,0,2) \pmod 3. \tag{E124.17}$$

Here ϕ is restriction to the radical β_3. The positive power-column sign is fixed by the principal idele relation $nI_i + \chi_3(a_i)e_3 + \chi_{53}(a_i)e_{53} = 0$. Reducing any idele by its ideal class and then its local unit components gives a surjection from this presentation to Γ. Its determinant is 2187, so the map is an isomorphism. These are the actual idele and Artin maps of Milne V Proposition5.2 and Theorems5.3 and5.5. Local norm compatibility follows from the proved cup-product Res/Cor identities in III Theorem3.1: in Tate degree -2 the group maps are transfer and inclusion, giving respectively inclusion/transfer and norm/restriction on Artin maps. No proof is attributed to III Lemma3.2, whose printed proof is omitted.

The full preimage of $\ker\phi$ in $\mathbb{Z}_3^5$ has basis

$$K = \begin{pmatrix} 3 & -1 & -1 & 0 & -2 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \text{relations } K^{-1}R. \tag{E124.18}$$

The certificate retains unimodular matrices U, V satisfying $U(K^{-1}R)V = \mathrm{diag}(27,9,3,1,1)$. The actual forward map is UK^{-1} on the kernel lattice; the inverse is KU^{-1} on Smith representatives. Complete ambient matrices, including thirds and the trivial Smith coordinates, are serialized and decoded exactly.

The actual relation $\mathfrak{q}_{3,2} = I_1^6 I_2^3 I_3(x_Q)$ has $\chi_{53}(x_Q) = 2$. Removing its three-valuation with the norm uniformizer $1+\beta_2$ gives

$$\chi_3(x_Q) = 6\cdot 1 + 3\cdot 0 + 1\cdot 2 - 2 = 0 \pmod 3. \tag{E124.19}$$

Thus the upper three-prime has kernel vector $(0,-2,6,3,1)$. Applying UK^{-1} gives $(21,0,1,0,0)$. Projection onto the last three idelic coordinates is the actual norm to D_2; it gives N_2 in (E124.3). The trace identity gives $j_2(b)=3a$ for every lift $N_2(a)=b$. The norm kernel has exponent three, so this rule is independent of the lift and gives the displayed j_2. The independent verifier checks the rule on every lift of each of the 243 lower classes, both trace compositions, all 729 upper classes, both kernels and both S-quotient maps.

Unit defects and completed kernels. An invariant ideal is a product of extended lower ideals and ramified primes: its coefficients are constant on each unramified orbit, and ramification gives the extra prime generator. The fifty-three prime is principal. Since N_2 is onto and its trace is three, the strongly ambiguous subgroup is $3D_3+\langle q_3\rangle$. Its order is 81, and the two-row quotient in (E124.5) has exactly this kernel.

The Hilbert–90 maps have literal inverses. Given an invariant class represented by a divisor A, choose x with $(\sigma_2-1)A=\operatorname{div}(x)$ and send it to Nx. This is an ordinary unit, well-defined modulo unit norms. Conversely, for $u=Nx\in E_2$, the divisor of x has norm zero and equals $(\sigma_2-1)A$. On a split orbit this is the difference description of the augmentation kernel; at a fixed prime the coefficient must be zero. Changing the norm witness by Hilbert90 changes A by a principal divisor and an invariant divisor. The latter is exactly the strongly ambiguous ambiguity. These maps are inverse, as in the ideal-level proof of Lemmermeyer Proposition1.

A finite-index subgroup of a finitely generated abelian group has its induced three-power topology cofinal with its own: an integer killing the quotient gives both inclusions of powers. Thus the unit norm sequence and its finite quotient remain exact after completion. The ranks at levels $d,d+1$ are $3^d-1,3^{d+1}-1$; lower cubes are unit norms, so the image has finite index and the free kernel rank is $2\cdot 3^d$. The literal root subgroups are μ_3, and norm is cubing, so the kernel torsion is exactly μ_3. Choose a basis of the kernel modulo torsion and lifts k_i. The displayed kernel isomorphism is $(z,(a_i))\mapsto z\prod_i k_i^{a_i}$; its inverse takes quotient coordinates, divides by this product and reads the remaining root. This specifies every basis and lift choice.

Later actual norms and bounded ordinary growth. Every N_d is onto. Indeed the ramified cyclic cubic L_{d+1}/L_d intersects the ordinary Hilbert three-class field of L_d only in L_d. Extend a prescribed Hilbert Frobenius to the compositum with trivial component on L_{d+1} and choose a prime with that Frobenius. A relative degree-one prime above it has the prescribed ideal norm. This uses the actual norm/Frobenius square of Milne V3.3, the Hilbert-field identification V3.6/Example3.9, and VIII7.4.

Total ramification gives $N_dq_{d+1}=q_d$ and $j_dq_d=3q_{d+1}$, proving $a_d\le a_{d+1}\le a_d+1$. The S-class norm is an isomorphism, so $\ker N_d\subset\langle q_{d+1}\rangle$ and has order at most three. Also $N_d\sigma_d=N_d$ and $\sigma_dq_{d+1}=q_{d+1}$. Consequently $T=\sigma_d-1$ has $T^2=3T=0$. Norm of an ideal followed by extension is the product of its conjugates, so $j_dN_d=1+\sigma_d+\sigma_d^2=3+3T+T^2=3$. The other composition is three as well.

Set $F=L_2$ and

$$F_\infty=F(\mu_{3^\infty},53^{1/3^\infty}). \tag{E124.20}$$

Each literal L_d is a subfield, through $\beta_d=53^{1/3^d}\zeta_{3^{d+1}}$ with the fixed roots. There is one place of F over three. Kato Theorem1.1(2) gives finite index for the image of the actual inverse system of completed three-units over all finite subfields of (E124.20) in $\mathcal{O}_F[1/3]^\times\otimes\mathbb{Z}_3$ [3]. Restriction to the coordinates L_d is an actual map of norm systems. Relative residue degree one makes the three-valuation unchanged. The finite-index image at F has nonzero three-valuation, since the whole valuation image there is $3\mathbb{Z}_3$. That nonzero value is divisible by every 3^{a_d}. Hence $h<\infty$. Kato supplies boundedness and coherent systems here, not a numerical value of its finite index.

Higher bases and inverse maps. The nondecreasing bounded integers a_d are eventually constant. Surjectivity and equal orders then make the ordinary norms isomorphisms. Their

inverse limit is that eventual finite group and maps onto D_3. Its coherent prime class q has order 3^h. In D_3 put $x_3 = (7,0,0)$ and $y_3 = (0,1,0)$; then $9x_3 = 3q_3$ and $9y_3 = 0$. Choose actual norm lifts x, y in the inverse group. Their S-quotient coordinates are a basis. Thus

$$9x = 3uq, \qquad 9y = 3vq, \qquad u \equiv 1 \pmod 3, \quad v \equiv 0 \pmod 3. \tag{E124.21}$$

Indeed their coefficients on q reduce modulo nine to three and zero under the actual projection to D_3. Choose representatives for u, v, and define

$$X = u^{-1}x, \qquad Y = y - vX, \qquad Z = q - 3X. \tag{E124.22}$$

Then $9X = 3q$, $9Y = 0$, $3Z = 0$ and $q = 3X + Z$. The three resulting cyclic groups surject onto the inverse class group and have total order $81 \cdot 3^h$, its order by (E124.14). Hence this is an isomorphism. An inverse reads quotient coordinates r, s modulo nine in $\overline{X}, \overline{Y}$, writes the residual class as tq, and returns $(r + 3t, s, t \bmod 3)$. Changing r by nine changes t by minus three, so the answer is unchanged. The other ambiguities are killed by $9Y = 0$ and the order of q.

Taking actual norms of X, Y, Z gives bases at every $d \geq 3$ with the same formulas and h replaced by a_d. The inverse construction commutes with norm, which proves the coordinate maps in (E124.6). Its extension rule follows from the trace identity and the norm kernel's exponent three. For all $d \geq 2$ the strongly ambiguous subgroup is therefore $3D_{d+1} + \langle q_{d+1} \rangle$ and has index nine. Lemmermeyer Theorem1(2) gives

$$[E_d : N_d E_{d+1}] = \frac{3|D_d|}{|D_{d+1}|/9} = 3^{\,3 - a_{d+1} + a_d}. \tag{E124.23}$$

All lower cubes are unit norms, proving the asserted elementary group type. The actual subgroup is still $N_d E_{d+1}$. For its field-norm part the same Hilbert–90 map uses $D_{d+1}^{\sigma_d}$ and quotients by the strongly ambiguous subgroup. This specifies the comparison even where the individual σ_d action has not been evaluated. The earlier rank and literal-root computation proves the remaining completed-kernel claim.

The coherent valuation image. Put $W_d = \mathcal{O}_{L_d}[1/3]^\times$. Its completed three-valuation image is exactly $3^{a_d}\mathbb{Z}_3$, by (E124.14). For $e \in \bigcap_d 3^{a_d}\mathbb{Z}_3$, every completed valuation fibre at level d is nonempty and compact. A finite chain of norm equations is solved by choosing in its highest fibre and taking norms. In the product of compact fibres, these closed equations have the finite intersection property. Their intersection proves both directions of

$$v_3\left(\varprojlim_{d,N} \widehat{W}_{d,3}\right) = \bigcap_d 3^{a_d}\mathbb{Z}_3 = 3^h\mathbb{Z}_3. \tag{E124.24}$$

Choose γ in this limit with valuation 3^h. The literal tuple $\boldsymbol{\beta}$ is norm coherent, has zero three-valuation and fifty-three valuation one. In any coherent S-unit tuple the latter valuation is constant. Divide by the indicated power of $\boldsymbol{\beta}$, then by the indicated power of γ; the result is a coherent ordinary unit. This proves (E124.8) and both compositions. Compact finite-level unit groups and continuous three-adic exponentiation also prove continuity of each inverse.

An ordinary unit has absolute norm ± 1, which is trivial in the three-power completion of $\mathbb{Q}^\times$. The two absolute residue degrees are one and two. These give (E124.9) on the actual norms and then (E124.10) using E.123's fixed divisor splitting. The numerical value of h and a literal coherent minimal-valuation tuple remain the specified next arithmetic step. All stage-three computations, boundedness, the exact image and the displayed comparison maps are proved without assigning that value. □

References

[1] J. S. Milne, *Class Field Theory*, version4.03, 2020.

[2] F. Lemmermeyer, *The Ambiguous Class Number Formula Revisited*, arXiv:1309.1071, 2013.

[3] K. Kato, *Universal Norms of p-Units in Some Non-Commutative Galois Extensions*, Documenta Mathematica, Extra Volume Coates (2006), 551–565.